

▼ **Figure 34.32** An emu (*Dromaius novaehollandiae*), a flightless bird native to Australia.



Several groups of living and extinct birds include one or more flightless species. The **ratites**, an order of birds that includes the ostrich, rhea, kiwi, cassowary, and emu, are all flightless (**Figure 34.32**). In ratites, the sternal keel is absent, and the pectoral muscles are small relative to those of birds that can fly.

Penguins make up another flightless order of birds, but, like flying birds, they have powerful pectoral muscles. They use these muscles to “fly” in the water: As they swim, they flap their flipper-like wings in a manner that resembles the flight stroke of a more typical bird (**Figure 34.33**). Certain species of rails, ducks, and pigeons are also flightless.

Although the demands of flight have rendered the general body forms of many flying birds similar to one another, experienced bird-watchers can distinguish species by their profile, colors, flying style, behavior, and beak shape. The skeleton of a hummingbird’s wing is

▼ **Figure 34.33** A king penguin (*Aptenodytes patagonicus*) “flying” underwater. With their streamlined shape and powerful pectoral muscles, penguins are fast and agile swimmers.



▼ **Figure 34.34** Hummingbird feeding while hovering. A hummingbird can rotate its wings in all directions, enabling it to hover and fly backward.



▼ **Figure 34.35** A specialized beak. This American flamingo (*Phoenicopterus ruber*) dips its beak into the water and strains out the food.



unique, making it the only bird that can hover and fly backward (**Figure 34.34**). Adult birds lack teeth, but during the course of avian evolution their beaks have taken on a variety of shapes suited to different diets. Some birds, such as parrots, have crushing beaks with which they can crack open hard nuts and seeds. Other birds, such as flamingoes, are filter feeders. Their beaks have “strainers” that enable them to capture food particles from the water (**Figure 34.35**). Foot structure, too, shows considerable variation. Various birds use their feet for perching on branches (**Figure 34.36**), grasping food, defense, swimming or walking, and even courtship (see Figure 24.3e).